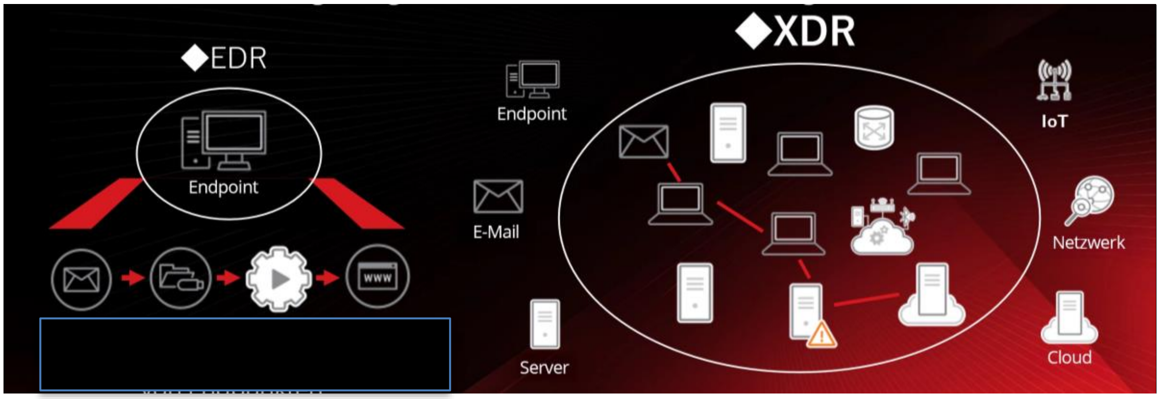
EDR and XDR Solutions Research & Comparison Report

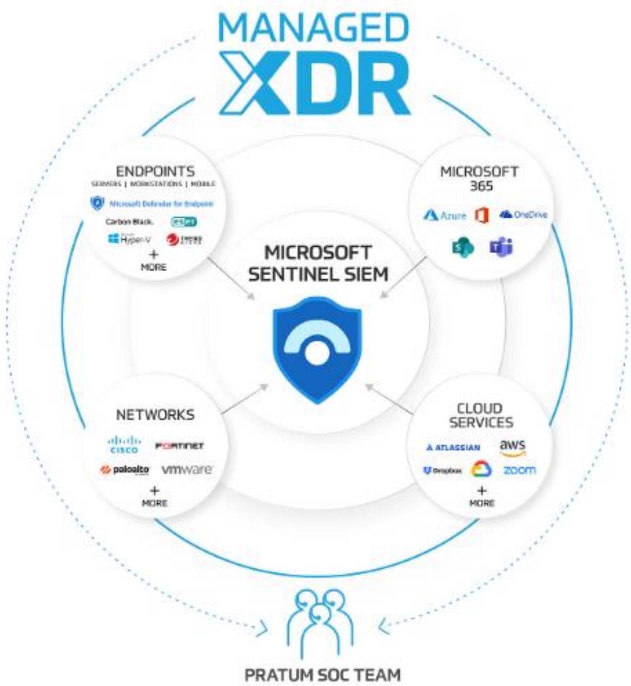


1. General Information

- **Assessment Date:6/5/2026**
- **Analyst Name:** Mustafa Jbili
- **Scope of Work:** Cybersecurity Analyst
- **Objective :**
  This report provides a professional overview of Endpoint Detection and Response (EDR) and Extended Detection and Response (XDR) solutions.
  It explains the evolution from EDR to XDR, compares their capabilities, and analyzes leading open-source and commercial solutions available in the cybersecurity market.

2. Introduction

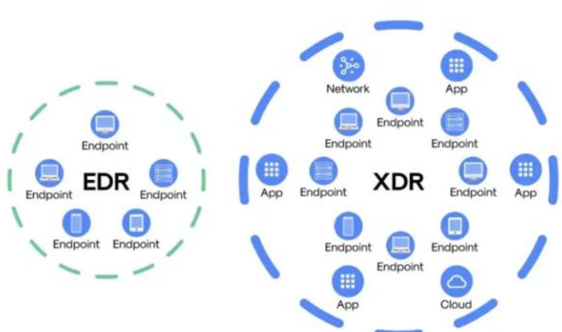
Endpoint Detection and Response (EDR) is a cybersecurity solution designed to monitor, detect, investigate, and respond to threats targeting endpoints such as laptops, servers, and workstations.
EDR solutions collect endpoint telemetry, analyze suspicious behavior, and enable security teams to investigate incidents in real time.



Extended Detection and Response (XDR) expands the capabilities of EDR by integrating telemetry from multiple security layers including endpoints, networks, cloud environments, email systems, identities, and applications.
XDR provides centralized visibility and correlated threat detection across the organization.

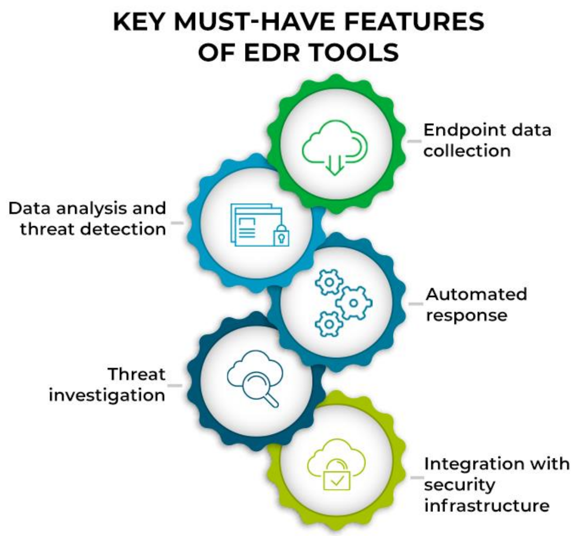
Evolution from EDR to XDR:

- Traditional antivirus focused mainly on signature-based malware detection.



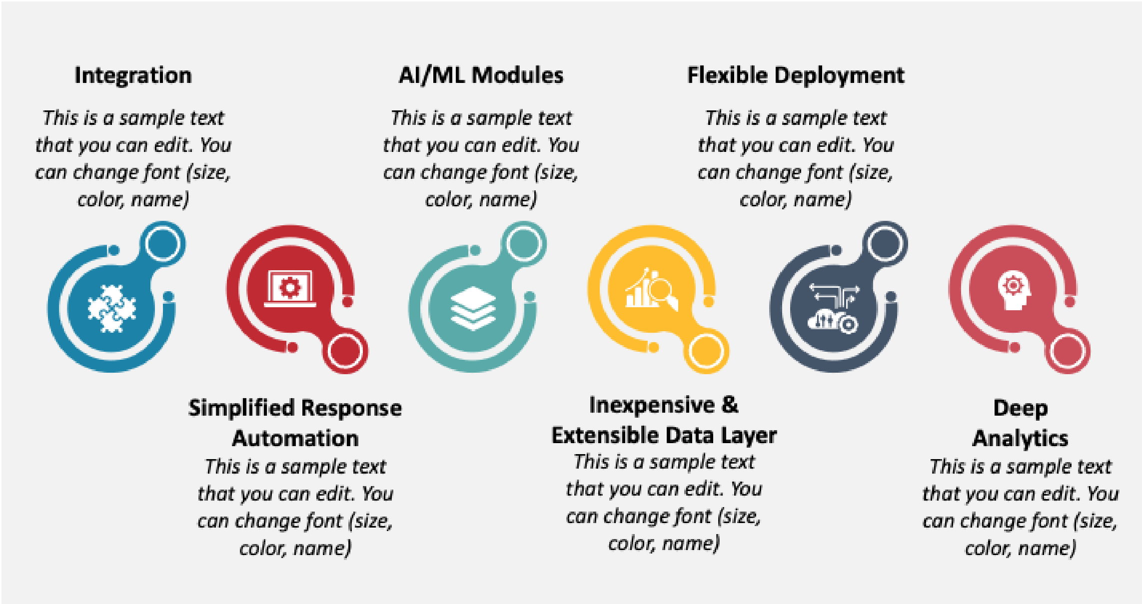
- EDR introduced behavioral analysis, threat hunting, and incident response for endpoints.
- XDR evolved further by correlating data from multiple sources to improve detection accuracy and automated response.

Key Components of EDR:



- Endpoint agents
- Threat detection engine
- Behavioral analytics
- Incident response tools
- Threat intelligence integration

Key Components of XDR:



- Unified telemetry collection
- Cross-platform analytics
- AI/ML-based detection
- Automated orchestration and response
- Integration with SIEM, SOAR, cloud, and identity tools

3. Comparison Criteria

The following criteria are commonly used to compare EDR and XDR solutions:

1. Scope of Visibility:
- EDR focuses primarily on endpoints.
- XDR extends visibility to cloud workloads, identities, email, networks, and applications.

2. Detection Capabilities:
- Signature-based detection
- Behavioral analytics
- AI/ML-powered anomaly detection
- Threat intelligence correlation

3. Response Capabilities:
- Automated endpoint isolation
- Process termination
- Rollback and remediation
- Cross-domain automated response

4. Integration:
- Compatibility with SIEM and SOAR platforms
- API availability
- Third-party integrations

5. Deployment and Management:
- Cloud-native vs on-premises
- Ease of deployment
- Centralized management dashboards

6. Scalability:
- Ability to support enterprise-scale environments
- Performance across hybrid infrastructures

7. Cost:
- Licensing model
- Infrastructure requirements
- Maintenance and operational costs

4. EDR Solutions

Free/Open-Source EDR Solutions:

1. Wazuh
- Open-source security monitoring and EDR platform.
- Features include log analysis, file integrity monitoring, vulnerability detection, and threat hunting.
- Strong integration with Elastic Stack.

2. Velociraptor
- Open-source digital forensics and incident response platform.
- Provides endpoint visibility and threat hunting capabilities.

3. OSQuery
- SQL-based endpoint visibility framework developed by Facebook.
- Useful for endpoint monitoring and investigations.

Commercial EDR Solutions:

1. CrowdStrike Falcon Insight
- Cloud-native EDR platform.
- Strong behavioral analytics and threat intelligence.
- Widely recognized for high detection accuracy.

2. Microsoft Defender for Endpoint
- Integrated with Microsoft ecosystem.
- Strong protection for Windows environments.
- Includes automated investigation and remediation.

3. SentinelOne Singularity
- AI-powered autonomous response.
- Excellent ransomware rollback capabilities.

4. Sophos Intercept X
- Combines EDR with anti-ransomware technology.
- User-friendly management interface.

5. Trend Micro Vision One
- Strong visibility and centralized management.
- Includes threat intelligence integration.

5. XDR Solutions

Commercial XDR Solutions:

1. Microsoft Defender XDR
- Integrates endpoint, email, identity, and cloud protection.
- Strong choice for Microsoft 365 environments.

2. Palo Alto Cortex XDR
- Correlates network, endpoint, and cloud telemetry.
- Advanced analytics and automation capabilities.

3. CrowdStrike Falcon XDR
- Extends Falcon platform across multiple security domains.
- Cloud-native architecture with advanced threat intelligence.

4. SentinelOne Singularity XDR
- AI-driven autonomous detection and response.
- Strong endpoint and cloud workload visibility.

5. Trend Micro Vision One
- Provides cross-layer detection and centralized incident management.

6. Cisco XDR
- Integrates Cisco security ecosystem with third-party tools.
- Focuses on automated investigation and response.

6. Analysis and Recommendations

When EDR is Sufficient:
- Small and medium-sized businesses with limited infrastructure.
- Organizations mainly concerned with endpoint protection.
- Companies with limited security budgets.

When XDR is Recommended:
- Enterprises with hybrid or cloud environments.
- Organizations requiring centralized security visibility.
- Security teams handling sophisticated multi-stage attacks.

Market Trends:
- Increasing use of AI and machine learning in detection.
- Growth of managed XDR (MXDR) services.
- Integration of identity and cloud telemetry into security platforms.
- Automation and orchestration becoming critical requirements.

Factors to Consider When Choosing a Solution:
- Organization size and infrastructure complexity.
- Existing security stack compatibility.

- Budget and licensing model.
- Internal security expertise.
- Compliance and regulatory requirements.

General Recommendations:

- Microsoft Defender XDR is suitable for organizations heavily invested in Microsoft services.
- CrowdStrike Falcon is ideal for advanced threat detection and threat intelligence.
- SentinelOne is strong for autonomous response and ransomware protection.
- Wazuh is an excellent low-cost/open-source option for learning and small deployments.

7. References

- SentinelOne Cybersecurity Research Articles (2026)
- Palo Alto Networks Cybersecurity Resources
- Microsoft Security Documentation
- CIOPages Buyer's Guide for EDR/XDR
- Community discussions from cybersecurity professionals and IT administrators

Feature	EDR	XDR
Scope	Endpoints only	Endpoints + Network + Cloud + Email + Identity
Data Sources	Endpoint telemetry	Multiple telemetry sources
Detection	Endpoint-focused	Cross-domain correlated detection
Response	Endpoint remediation	Organization-wide response
Complexity	Lower	Higher
Cost	Generally lower	Generally higher